

TOKEN BALANCING, UPDATE SPARSITY, AND SUPERVISION DENSITY IN MULTI-TEACHER ON-POLICY DISTILLATION

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ABSTRACT

Multi-teacher on-policy distillation (MOPD) consolidates specialized capabilities into a shared student, but how supervision choices translate into parameter changes and capability gains remains unclear. We study loss averaging, update geometry, and vocabulary supervision by combining online capability evaluation with interventions at matched student and optimizer states. First, balancing domain weights leaves a separate choice of response weighting, which changes raw-gradient directions. Shared Adam history makes the resulting local updates more aligned, while online differences depend on the supervision objective and optimizer. Second, broad FP32 parameter changes coexist with sparse BF16-visible changes. Subtracting the zero-current-gradient step removes most alignment between teachers on their respective domain inputs, showing that apparent agreement can reflect shared optimizer history. Third, broader vocabulary feedback produces similar local update concentration in the measured high-coverage regime, but its capability effects vary across domains and checkpoints. Additional-domain testing also shows task-dependent trade-offs, and retention-constrained selection yields the same checkpoints as unconstrained development utility. Together, these findings show why loss weights, optimizer state, and numerical precision must be considered alongside per-domain capability when evaluating MOPD design choices.

1 INTRODUCTION

Multi-teacher on-policy distillation is increasingly used in LLM post-training to consolidate specialized capabilities into a single model (Xiaomi LLM-Core Team, 2026; NVIDIA, 2026; Kimi Team et al., 2026; DeepSeek-AI et al., 2026). In this paradigm, specialized teachers provide token-level feedback on responses generated by the student itself. The goal is to integrate their expertise while retaining the student’s existing capabilities, but these outcomes need to be assessed separately across domains.

Recent work has advanced both the practice and analysis of this approach. Ma et al. (2026) demonstrate capability integration using policy-gradient and top- k distillation losses. Open-MOPD identifies uneven capability transfer and develops mechanisms for balancing domain token shares, allocating supervision according to student–teacher gaps, and refreshing student-dependent rewards (Gao et al., 2026). The Nemotron 3 Ultra report further finds that top- k and full-vocabulary matching do not outperform sampled-token supervision in its preliminary experiments (NVIDIA, 2026, Section 3.3.5). Complementary studies characterize the sparsity and geometry of OPD parameter changes, including their concentration at checkpoint precision and their evolution during training (Yu et al., 2026; Shen et al., 2026). These findings motivate examining how supervision choices translate into parameter changes and capability outcomes in a shared student.

Several distinctions are central to this examination. Balancing domain weights still leaves open how responses of different lengths are weighted within each domain. Concentrated changes in stored model weights do not, by themselves, describe the breadth of optimizer updates. Likewise, supervising more vocabulary alternatives need not produce a proportional increase in changed parameters, and similar local update concentration may accompany different capability outcomes. We therefore

ask: *how do loss averaging and vocabulary supervision affect learning in MOPD, and what do parameter-change measurements reveal about these choices?*

We combine algebraic analysis of loss averaging, local interventions at fixed student and optimizer states, and capability evaluations along training. The interventions isolate how a design choice changes a local update; online comparisons measure the capability outcomes as each student’s responses and optimizer history evolve. Our contributions are threefold:

1. **Separating domain balancing from response weighting (Section 3).** We distinguish global-token, domain-token, and domain-response averaging and show that within-domain length weighting changes raw-gradient direction even with identical domain weights. Clipping preserves these directions, whereas shared Adam history makes the local updates substantially more similar. Similar final mean scores under top-64 supervision coexist with different capability trajectories; the expanded objective–optimizer comparison limits how broadly this endpoint pattern can be generalized.
2. **Separating current teacher signals from history and precision effects (Section 4).** We show that broad FP32 changes coexist with sparse BF16-visible changes, and that teacher-update alignment is sensitive to shared optimizer history. Subtracting the step produced with zero current gradient removes most alignment under domain-routed inputs. Update overlap and direction therefore need to be interpreted with both numerical representation and optimizer state specified.
3. **Comparing vocabulary supervision through both updates and capability (Section 5).** Sampled-token, top- k , and full-vocabulary losses produce similar local update concentration in the measured high-coverage regime. Online sampled-token, top-16, and top-64 comparisons nevertheless show domain- and checkpoint-dependent trade-offs, with no uniform gain from broader feedback. Local concentration alone is therefore insufficient to rank supervision choices by capability.

Section 6 draws practical implications for weighting, update diagnostics, and checkpoint selection. An additional-domain test provides a separate capability comparison; retention-constrained and unconstrained utility selection choose the same checkpoints in the tested trajectories, so these results do not establish an added benefit from the constraint.

2 PRELIMINARIES

Multi-teacher on-policy distillation (MOPD). For domain d , the student samples a response $y \sim p_\theta(\cdot \mid x)$ to a prompt $x \sim \mathcal{D}_d$. The assigned teacher q_d provides supervision at the student prefixes $h_t = (x, y_{<t})$. Following the distinction between token-level supervision and loss aggregation in Open-MOPD (Gao et al., 2026), we specify these two choices separately. For a rollout batch $\mathcal{B}$, let T_i be response i ’s valid-token count and $\ell_i = T_i^{-1} \sum_t \ell_{i,t}$ its mean token loss, with the sum over valid positions. The training objective on this batch is

$$\mathcal{L}(\theta) = \sum_{i \in \mathcal{B}} w_i \ell_i(\theta), \quad w_i \geq 0, \quad \sum_{i \in \mathcal{B}} w_i = 1. \quad (1)$$

The token loss specifies the teacher feedback, while w_i specifies how responses contribute to the batch objective. Writing ℓ_i as a response mean leaves the averaging rule open: w_i can depend on response length and domain, as defined below. Training holds sampled responses, masks, and averaging weights fixed when differentiating; fresh responses are generated at each update.

Sampled-token, top- k , and full-vocabulary supervision. These choices determine which vocabulary alternatives receive teacher feedback at each prefix. Write $p_\theta(v) = p_\theta(v \mid h_t)$ and $q_d(v) = q_d(v \mid h_t)$. Full-vocabulary supervision computes reverse KL over the vocabulary $\mathcal{V}$:

$$\ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_\theta(v) \log \frac{p_\theta(v)}{q_d(v)}. \quad (2)$$

Sampled-token supervision (PG) uses only the generated token $a = y_t$:

$$\ell_{\text{PG}}(a) = -\text{sg}[\log q_d(a) - \log p_\theta(a)] \log p_\theta(a), \quad a \sim p_\theta. \quad (3)$$

Here sg stops gradients through the log-ratio coefficient. At a fixed prefix, averaging this gradient over $a \sim p_\theta$ recovers the full-vocabulary reverse-KL gradient (Appendix D).

Top- k supervision uses the k most probable tokens under the student or teacher. We use their intersection: let S and T be the student and teacher top- k sets fixed at rollout, and $I = S \cap T$. The loss is

$$\ell_{\text{top-}k} = - \sum_{v \in I} \text{sg} \left[\frac{p_\theta(v)}{\sum_{u \in S} p_\theta(u)} (\log q_d(v) - \log p_\theta(v)) \right] \log p_\theta(v). \quad (4)$$

The probabilities use full-vocabulary normalization; the detached weights are normalized over S before restricting to I . An empty intersection gives zero loss. We call the number of supervised alternatives *supervision density*.

Loss averaging. We compare three choices of w_i in Equation 1. Let $\mathcal{B}_d$ contain the responses from domain d , and let $\lambda_d \geq 0$ be the target domain weights with $\sum_d \lambda_d = 1$. We consider batches of nonempty responses with every domain represented. For $i \in \mathcal{B}_d$,

$$w_i = \begin{cases} \frac{T_i}{\sum_{j \in \mathcal{B}} T_j}, & \text{global-token (GT),} \\ \frac{\lambda_d T_i}{\sum_{j \in \mathcal{B}_d} T_j}, & \text{domain-token (DT),} \\ \frac{\lambda_d}{|\mathcal{B}_d|}, & \text{domain-response (DR).} \end{cases} \quad (5)$$

Under GT, T_i cancels the response-mean denominator in ℓ_i , so every valid token receives equal weight and each domain’s total weight equals its observed token share. DT fixes each domain’s total weight to λ_d while retaining equal token weights within that domain. DR distributes λ_d equally across responses within each domain. Thus all three rules instantiate Equation 1 with the same token loss but different response weights.

Update alignment. Let u_d be the parameter change proposed by an optimizer using domain d ’s mean loss at a given student and optimizer state. For nonzero updates, define $C_{de} = (u_d^\top u_e) / (\|u_d\| \|u_e\|)$. Negative values indicate opposing update directions; positive values indicate alignment. This geometric measure does not by itself determine the effect on domain capabilities.

3 TOKEN BALANCING AND RESPONSE LENGTH

3.1 WHY CAN TOKEN BALANCING CHANGE MOPD OPTIMIZATION?

Equal prompt quotas do not imply equal optimization influence in MOPD: domains and teachers elicit responses of different lengths. The averaging rule therefore chooses what is balanced. GT weights every valid supervised token equally and lets long-response domains contribute more; DT fixes each domain’s total weight but still favors longer responses within a domain; DR also weights responses equally within each domain. These choices can change which capabilities dominate a batch, even when the prompt mixture is unchanged.

The distinction is visible directly in the gradient. Writing $g_i = \nabla_\theta \ell_i$, the token-weighted gradient within domain d satisfies

$$\frac{\sum_{i \in \mathcal{B}_d} T_i g_i}{\sum_{i \in \mathcal{B}_d} T_i} = \bar{g}_d + \frac{\text{Cov}_d(T, g)}{\bar{T}_d}. \quad (6)$$

The covariance term is nonzero whenever response length and gradient are related. DT retains this term; DR removes it. Correcting GT by the target-to-observed domain token share gives DT exactly, as in Open-MOPD (Gao et al., 2026, Section 4.1, Eq. (8)). We therefore test two linked questions: whether length weighting changes raw gradients and capability balance, and whether clipping and Adam preserve or mask that difference before it reaches the parameters. Appendix B gives the derivation.

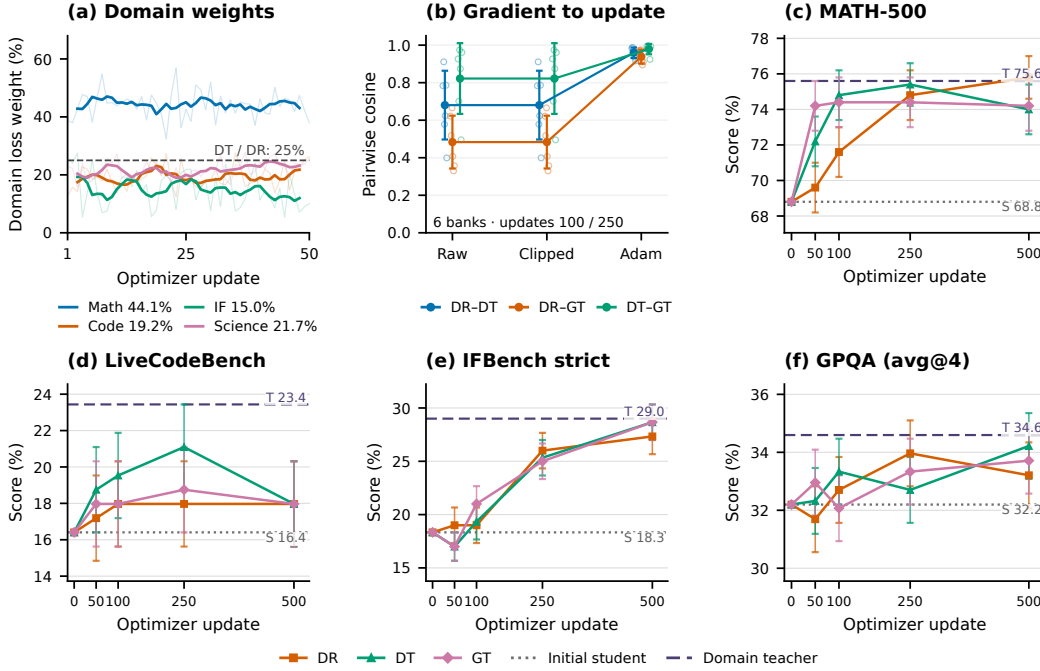


Figure 1: Loss weighting, optimizer mediation, and capability. (a) GT domain weights: faint lines show batches, bold lines five-batch averages, and legend values overall means. (b) Pairwise cosines before/after clipping and after Adam; open points show banks, error bars mean $\pm$ SD. (c–f) GT/DT/DR task scores (mean $\pm$ SD), with initial-student (S) and teacher (T) references. Capability summaries use the three reported experimental seeds; see Appendix F.

3.2 EXPERIMENT: AVERAGING RULES AND CAPABILITY BALANCE

Comparison. We use a Qwen3-1.7B student and four frozen domain teachers for mathematics, code, instruction following, and science. Three online branches use student–teacher top-64 intersection supervision (I64) and differ only in GT, DT, or DR averaging. They share initialization, teachers, prompt schedule, and optimizer settings; each update uses 16 prompts per domain. Capability is evaluated on MATH-500, a fixed LiveCodeBench subset, IFBench, and GPQA Diamond through update 500, with equal domain weights in the reported mean. Appendix A gives the training and scoring protocol.

Different paths lead to similar I64 endpoint means. Mathematics and instruction following each receive 25% of prompts but contribute 44.05% and 14.98% of GT’s valid tokens over the first 50 batches; GT therefore gives mathematics nearly three times the weight. DT and DR fix both domains at 25% (Figure 1a). This imbalance changes early trajectories, yet no rule wins consistently across seeds (Figure 1). At update 500, four-domain means are $38.58 \pm 0.75\%$ (DR), $38.71 \pm 0.80\%$ (DT), and $38.64 \pm 0.80\%$ (GT). The 0.14-point span is below the 0.75–0.80-point seed standard deviations. MATH and IFBench approach their teachers, LiveCodeBench retains a gap, and GPQA starts near its teacher; rule ranks change across tasks and checkpoints. Here token balancing changes capability trajectories while yielding similar final means.

Endpoint behavior depends on the objective and optimizer. Table 1 extends the comparison to PG under all three averaging rules, I16 under DR, and three PG/SGD runs. These endpoints use seed 42 and are separate from the three-seed normalization summaries above. All ten configurations improve the initial four-domain mean, from 33.93% to 38.03–39.46%. The averaging-rule spread is 0.33 points for PG/Adam and 1.07 points for PG/SGD, compared with 0.14 points for I64/Adam. Thus the narrow I64/Adam spread does not establish the same behavior across configurations. The

Table 1: Joint-training endpoint scores (%). All configurations improve the initial four-domain mean, with task-dependent differences across objectives and averaging rules. Mean weights domains equally.

Configuration	MATH-500	Code	IF	GPQA	Mean
Initial student	68.80	16.41	18.33	32.20	33.93
<i>Adam</i>					
PG-DR	73.20	17.97	27.00	34.97	38.29
PG-DT	73.20	17.97	27.00	33.96	38.03
PG-GT	76.20	17.97	27.33	31.94	38.36
I16-DR	75.40	18.75	27.00	33.46	38.65
I64-DR	75.80	17.97	27.33	33.21	38.58
I64-DT	74.00	17.97	28.67	34.22	38.71
I64-GT	74.20	17.97	28.67	33.71	38.64
<i>SGD</i>					
PG-DR	76.40	19.53	28.33	33.59	39.46
PG-DT	75.20	20.31	28.33	32.70	39.14
PG-GT	75.00	19.53	25.33	33.71	38.39

SGD learning rate and deployment differ from Adam, so this comparison does not isolate a single optimizer mechanism or establish an efficiency ranking. Appendix H reports the independent test.

The optimizer masks part of the gradient difference. On six fixed response banks, the mean raw-gradient cosines are 0.680 (DR–DT), 0.483 (DR–GT), and 0.822 (DT–GT); hence equal domain weights do not make DR and DT equivalent. Global clipping leaves every cosine unchanged, even though it acts very differently online: DR is clipped on all 500 updates and retains a median 9.3% of its gradient, compared with clipping on 46.2% of DT updates and 15.0% of GT updates (Figure 2a). Applying one common saved Adam state then raises the three FP32 update cosines to 0.960, 0.939, and 0.979. Adam history therefore changes the effective update rather than merely transmitting the current clipped gradient. Figure 2(b) shows that it also sets the step scale.

Shared history attenuates local differences. With saved-state Adam, the three reductions’ immediate held-out KL changes are close relative to bank variation (Figure 2c). Same-norm SGD exposes a wider, checkpoint-dependent spread: all three mean changes improve at update 100 but worsen at update 250. Clipping changes gradient magnitudes but not angles, whereas shared Adam history pulls distinct gradients toward similar updates. These interventions identify a local mechanism; they do not explain the full online trajectories, whose responses and optimizer histories evolve separately. Appendix B.1 gives the intervention details and raw-gradient comparisons; Figure 6 reports paired capability contrasts.

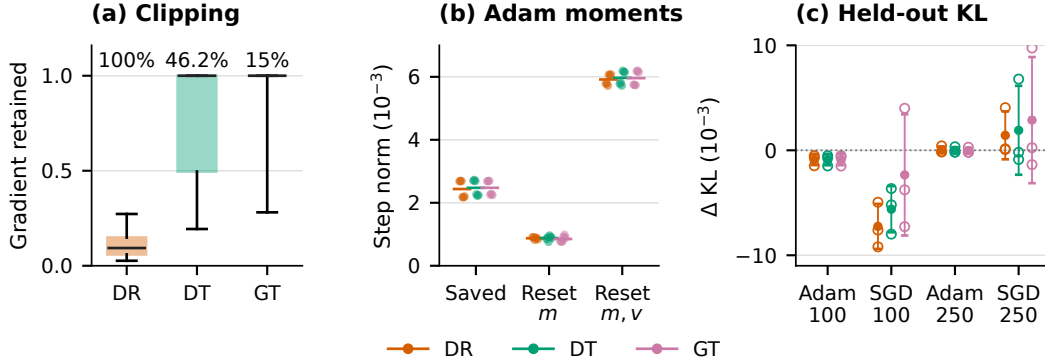


Figure 2: Averaging rules differ in clipping, while Adam history shapes step size and immediate loss changes. (a) Gradient fraction retained; labels give clipping rates, boxes the interquartile range, and whiskers the 5th–95th percentiles. (b) FP32 step norms; bars show bank means. (c) Held-out reverse-KL changes; lower is better. Open points denote banks; filled points and error bars show mean $\pm$ sample SD.

4 UPDATE SPARSITY AND TEACHER OVERLAP

This study examines how numerical representation and optimizer history affect the concentration, overlap, and direction of teacher-conditioned parameter changes. Prior work observes concentrated changes under dense distillation supervision (Yu et al., 2026). We examine the corresponding measurements in a jointly trained student.

4.1 HOW CONCENTRATED ARE JOINT PARAMETER CHANGES?

Write $\Delta = \theta - \theta_0$ for cumulative changes from initialization and $u = \theta_{\text{after}} - \theta_{\text{before}}$ for an individual optimizer step. For either change vector z with P coordinates, we report the fraction of parameters whose absolute change exceeds threshold τ :

$$f_{\tau}(z) = \frac{1}{P} \sum_{j=1}^P \mathbf{1}\{|z_j| > \tau\}. \quad (7)$$

We also report the smallest coordinate fraction accounting for 90% of the squared change. This measures concentration without choosing an absolute threshold. Both quantities are computed separately for FP32 optimizer weights and BF16 stored model weights.

Joint-training comparison. The joint PG and I64-DR branches share the training settings in Appendix A, including domain-response averaging. At update 500, PG changes 7.30% of BF16 coordinates, with 90% of squared change in 3.79% of coordinates; I64 changes 8.46%, with 90% of squared change in 4.38% (Figure 3a). Both joint runs concentrate their stored parameter changes in a small fraction of the model.

The corresponding FP32 nonzero fractions are 97.37% for PG and 97.17% for I64; their 90%-energy fractions are 34.03% and 35.14%. The optimizer steps at this checkpoint change 86.80% and 87.05% of FP32 coordinates, respectively. Thus sparse BF16-visible changes coexist with broad FP32 activity in both cumulative changes and individual steps. Appendix Figure 10 gives the trajectories, and Appendix C provides threshold and optimizer-state controls.

4.2 DO TEACHERS CHANGE THE SAME PARAMETERS IN ONE STUDENT?

Comparing teacher-specific updates requires identical copies of the joint student and its optimizer state. For teacher d , let $S_d(\rho)$ contain the largest-magnitude ρ fraction of its parameter changes. We

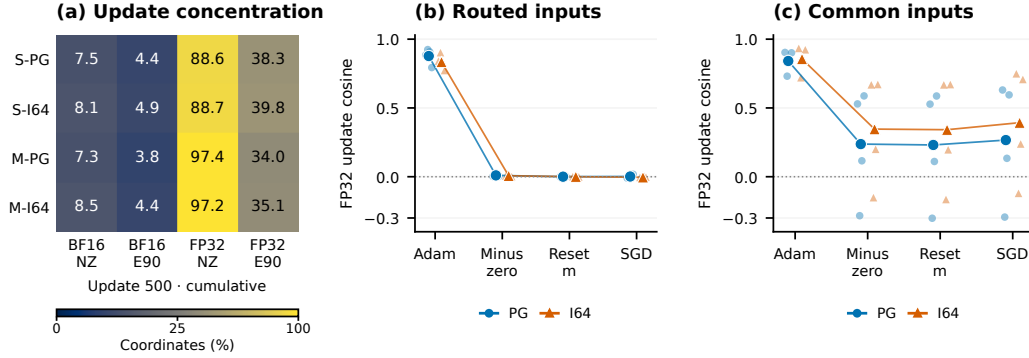


Figure 3: Stored BF16 changes are concentrated; removing the history-only step eliminates most routed teacher-update alignment. (a) Nonzero (NZ) and 90%-energy (E90) coordinate fractions; S/M denote single-teacher/joint training. Color uses a square-root scale. (b,c) FP32 teacher-update cosines for routed/common inputs. “Minus zero” denotes $U(g; H) - U(0; H)$. Small points show bank means over teacher pairs; large points average banks.

compare selected coordinates using Jaccard overlap:

$$J(d, e) = \frac{|S_d(\rho) \cap S_e(\rho)|}{|S_d(\rho) \cup S_e(\rho)|}. \quad (8)$$

Equal selection sizes make the teacher pairs comparable. Overlap describes locations; the signs of the changes determine whether contributions at those locations agree. This distinction follows the support-overlap analysis of Yu et al. (2026).

At the common M-PG/500 state, we compare teachers on shared prefixes (common inputs) and on their respective domain prefixes (routed inputs). Each result averages the six teacher pairs within each of four response banks, then averages banks; these are local diagnostic repeats. Routed teacher updates have mean FP32 cosines of 0.8774 for PG and 0.8347 for I64 under saved-state Adam. Resetting the first moment or using norm-matched SGD brings their mean cosines close to zero (Figure 3b). Appendix H.5 gives the joint overlap–direction and teacher-JS views, and Appendix C.2 gives raw-gradient input controls.

4.3 HOW MUCH ALIGNMENT REMAINS AFTER REMOVING THE HISTORY-ONLY STEP?

Let $U(g; H)$ denote an optimizer step for current gradient g at saved optimizer state H . We compare the complete teacher-conditioned step with its difference from the zero-current-gradient step,

$$\delta U_d = U(g_d; H) - U(0; H). \quad (9)$$

This difference retains the same parameters and optimizer state in both branches. It is a nonlinear counterfactual difference, not an additive decomposition of teacher contributions.

Under routed inputs, the mean PG/I64 cosines fall to 0.0098/0.0067 after this subtraction (Figure 3b). Under common inputs, they fall from 0.8415/0.8562 to 0.2373/0.3462 (panel c). The strong alignment of the complete steps therefore depends on shared optimizer history and input conditions. Coordinate overlap can remain substantial even when directions differ, and neither statistic establishes the absence of cross-domain capability harm. That question requires measuring how teacher-conditioned steps affect per-domain losses or capabilities.

5 VOCABULARY SUPERVISION: UPDATES AND CAPABILITY

Broader vocabulary supervision can change the gradient at a prefix without proportionally increasing the number of parameters changed by the optimizer. Dense supervision and concentrated checkpoint changes coexist in the experiments of Yu et al. (2026). We examine this relationship through a local comparison at a common state, online PG/I64 comparisons in single-teacher and joint training, and a joint I16 branch that varies the intersection support size.

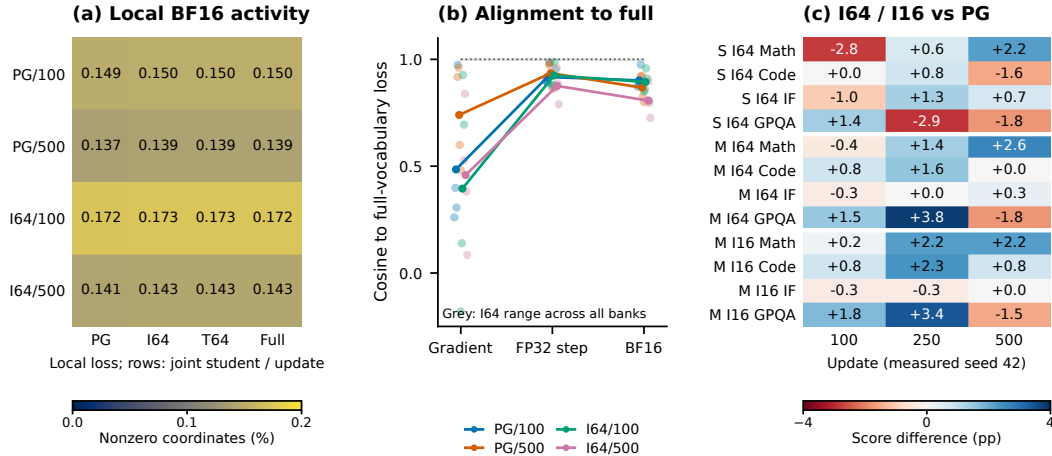


Figure 4: Vocabulary feedback changes gradient direction while local BF16 activity remains similar; capability gains vary by task and checkpoint. (a) Mean BF16 nonzero fractions by student state and loss. (b) Alignment with full-vocabulary supervision: PG bank means (lines) and individual banks (points), with the I64 range shaded. (c) I64/I16 score differences from PG within single-teacher (S) or joint (M) training. Positive values favor I64 or I16.

5.1 COMPARISON AND MEASUREMENTS

PG and I64 use Equations 3 and 4. The online configurations are S-PG and S-I64 with the mathematics teacher, and M-PG and M-I64-DR with four routed teachers. Within each setting, the branches share initialization, teachers, domain-response averaging, optimizer settings, and prompt schedule. The single-teacher branches use 64 mathematics responses per update; joint branches use 16 per domain. We compare PG and I64 within each setting at common checkpoints. The added joint I16 branch uses the same intersection definition and DR averaging, with $k = 16$ rather than 64. All added comparisons use seed 42.

The local experiment evaluates PG, I64, corrected teacher-top-64 supervision (T64), and full-vocabulary reverse KL at four joint student states: PG and I64-DR at updates 100 and 500. Within each state, all losses use identical prefix weights and the same saved optimizer state, with four banks totaling 96 sampled prefixes. We measure the fraction of changed parameters and the squared step norm carried by the largest 1% of coordinates. Appendix D gives the bank construction and T64 definition.

5.2 UPDATE CONCENTRATION

Local changes are similarly concentrated across losses. At M-PG/100, all four losses change about 88.0% of FP32 coordinates (Table 6). Mean BF16 nonzero fractions are 0.149% for PG and 0.150% for I64, T64, and full vocabulary. The largest 1% of coordinates carry about 8.8% of the FP32 squared step norm and all of the BF16 squared step norm. Local concentration is therefore very similar across these losses, as are the BF16 nonzero fractions at the other measured states (Figure 4a). Shared history contributes to this result: a zero-current-gradient control alone changes 0.140% of BF16 coordinates at M-PG/100.

Across these four states, the actual teacher–student intersection contains more than 99.9% of student probability on average, and none of the 384 sampled prefixes falls below 95%. The local comparison therefore describes a high-coverage regime. Table 10 gives the mass, support size, and retained weight ratio by state; Appendix C provides threshold and optimizer-state controls.

In a separate fixed-prefix sampling control, increasing PG draws from 1 to 16 to 64 raises mean gradient cosine with full vocabulary from 0.537 to 0.865 to 0.967 (Figure 5a). This supports sampling variability as one source of their local difference. Separate zero-gradient and moment-reset controls show how optimizer history affects BF16-visible changes (panels b,c).

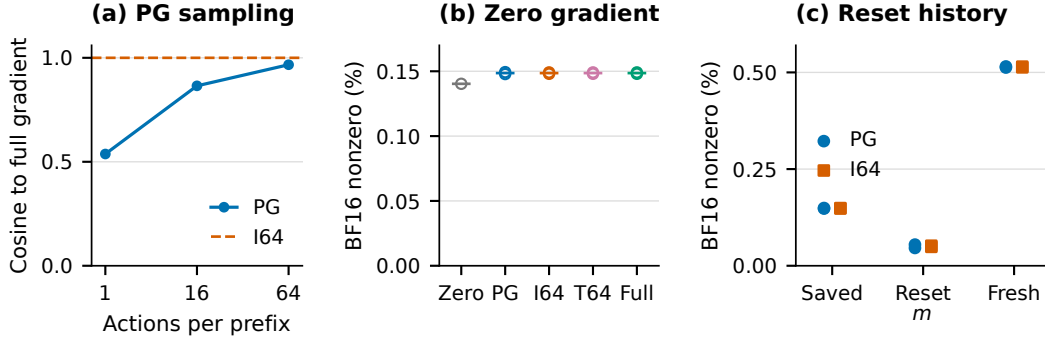


Figure 5: More PG actions improve gradient alignment, while optimizer history strongly affects BF16 activity. (a) Mean gradient cosine to full vocabulary; the dashed line denotes I64. (b) A zero-current-gradient step alone produces nearly as much BF16 activity as the supervised steps; bars show means. (c) Activity after resetting optimizer history. Points in (b,c) denote banks.

Joint trajectories accumulate different changes. Despite similar local concentration, joint I64 training accumulates broader BF16-visible changes than PG (Section 4; Figure 10). The local experiment fixes the student, prefixes, and optimizer state; the online comparison includes the feedback between each loss and the responses its student generates. Similar local concentration therefore does not imply identical training trajectories.

5.3 CAPABILITY UNDER SINGLE-TEACHER AND JOINT TRAINING

Figure 4(c) summarizes task-wise differences from PG. In single-teacher training, PG’s mathematics score exceeds I64 by 2.8 points at update 100, whereas I64 exceeds PG by 2.2 points at update 500. At the endpoint, I64 also scores higher on instruction following, while PG scores higher on code and GPQA. Appendix Table 2 gives the single-teacher and intermediate-checkpoint scores.

At the joint-training endpoint, I64-minus-PG differences are +2.60 points on mathematics, 0.00 on code, +0.33 on instruction following, and -1.77 on GPQA (Table 1). All four nominal paired 95% intervals include zero (Table 4). Earlier, at update 250, I64 scores higher in three domains and ties on instruction following; its mean advantage shrinks from 1.69 to 0.29 points by update 500. The apparent advantage therefore depends on the domain and checkpoint.

Increasing the intersection size does not yield uniform gains. Joint I16 and I64 have endpoint means of 38.65% and 38.58%, respectively (Table 1). Their additional-domain test means are 44.56% and 44.46%, with different per-domain trade-offs (Table 8). These point estimates do not establish equivalence or a general advantage for either support size. Changing k changes both the candidate set and its normalization; it is not a pure intervention on supervision density. I16 is an online capability comparison and is not included in the local geometry panels.

Broader feedback yields nearly unchanged concentration at matched states in the measured regime, while the trained students develop different capability profiles. Both measurements are needed to assess supervision choices.

6 PRACTICAL IMPLICATIONS AND CHECKPOINT SELECTION

The three studies support practical choices about weighting, diagnostics, and evaluation. They do not identify one configuration that is best across domains and evaluation sets.

Specify the intended weighting and measure the resulting capability profile. GT weights tokens equally; DT fixes domain weights while retaining length weighting within each domain; DR also weights responses equally. These are different objectives, even with the same prompt quotas.

Likewise, the PG/I16/I64 results favor selecting vocabulary feedback from per-domain capability and measured cost, with retained probability mass used to interpret coverage. The present runs do not provide a matched-cost efficiency ranking.

Interpret update geometry with its history and precision. Report the parameter representation, distinguish individual steps from cumulative changes, and compare teacher-conditioned updates at a common state. A zero-current-gradient reference helps assess how much alignment depends on shared history. Evidence of overlap or opposing directions should be followed by per-domain loss or capability measurements before inferring functional interference.

Evaluate checkpoint selection on separate questions. To test domain retention as a selection criterion, let $s_d(\theta)$ be a development score, with higher values better, and let $\epsilon_d \geq 0$ be a prespecified tolerance relative to initialization. Eligible checkpoints satisfy

$$s_d(\theta) \geq s_d(\theta_0) - \epsilon_d \quad \text{for every domain } d. \quad (10)$$

We select the eligible checkpoint with maximum equal-domain mean, retaining the initial student if none improves it by at least one percentage point. We compare this rule with unconstrained maximum utility and the final checkpoint, using a two-point domain tolerance fixed before the additional test results. Appendix H.2 gives the development grids and selection outcomes.

The additional test uses separately frozen questions from four domain sources, with 256 questions per domain and a distinct evaluation protocol (Appendix H.1). Relative rankings can change: PG-DR/SGD exceeds Adam by 1.18 mean points on the original benchmarks, but is 0.07 points lower on this test. Retention and unconstrained maximum utility select the same checkpoint on all ten trajectories: update 500 in nine cases and update 250 for PG-DT/Adam. In the latter case, the selected checkpoint scores 43.14% on the additional test, versus 43.51% at update 500. The retention constraint therefore has no demonstrated selection benefit here. Because the development benchmarks had been observed previously, this selection study remains exploratory.

7 RELATED WORK

MOPD combines specialized teachers through feedback on student-generated responses (Ma et al., 2026). Open-MOPD studies capability imbalance and introduces token-share balancing, gap-based weighting, and reward refresh (Gao et al., 2026). Its token-share correction maps directly to domain-token averaging under the same batch and target domain weights. Our first study separates this between-domain correction from within-domain response weighting and compares their capability outcomes.

Mukherjee et al. (2025) investigate small parameter subsets that support reinforcement-learning fine-tuning. Yu et al. (2026) find concentrated checkpoint changes under dense distillation supervision and overlap between the changes produced by different teachers in separately trained students. Our second question concerns teachers acting from the same joint student and optimizer state. Our third compares vocabulary supervision through both parameter concentration and capability. Shen et al. (2026) study OPD parameter geometry and select response positions for learning; their position selection concerns a different supervision dimension from the vocabulary alternatives scored at each prefix here.

8 DISCUSSION

The experiments use one Qwen3-1.7B student and four teachers derived from its initial model. The normalization capability summaries report seeds 42–44; the expanded objective/optimizer matrix and additional-domain test use seed 42. Local banks are diagnostic repeats, and question-bootstrap intervals measure evaluation uncertainty at fixed checkpoints. Neither substitutes for independent training runs. The local supervision comparisons also have high candidate coverage, limiting what they establish about vocabulary tails or less similar teachers. Appendix H.4 reports sensitivity to four known overlapping mathematics questions; this does not constitute exhaustive decontamination.

Local update geometry and long-term capability transfer remain distinct measurements. Fixed-state interventions reveal how weighting, history, and precision affect an update; online and additional-

domain evaluations establish the resulting capability trade-offs. Together they support conditional guidance for MOPD design, while broader claims about stability or efficiency require repeated training under matched budgets.

REPRODUCIBILITY STATEMENT

The accompanying evidence bundle contains figure and table values, configuration records, source hashes, scoring audits, and CPU plotting scripts. Appendix A gives the training data sources, evaluation protocol, checkpoint coverage, and uncertainty procedure. Appendix C specifies parameter measurements and local optimizer controls; Appendix H describes the additional-domain test, frozen checkpoint selections, and overlap sensitivity analysis.

ETHICS STATEMENT

The experiments use public model families and benchmark datasets to study optimization and capability transfer. We report standard benchmark scores with the stated evaluation denominators. Generative AI assisted manuscript editing and analysis code; the authors are responsible for the reported measurements and their interpretation.

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A EXPERIMENTAL PROTOCOL AND CAPABILITY RESULTS

The experiments compare averaging rules, teacher-conditioned updates, and vocabulary supervision using a shared student initialization and four domain teachers. The ten joint-training configurations in Table 1 include the original PG-DR and I64-DR/DT/GT branches. Mathematics-only PG/I64 runs provide the single-teacher comparison. All checkpoint axes count optimizer updates, and each trajectory comparison uses checkpoints completed for every included branch.

A.1 MODEL, DATA, AND TRAINING PROTOCOL

The student is Qwen3-1.7B (Yang et al., 2025). The four frozen teachers are mathematics, code, instruction-following, and science RL descendants of the same initial model. Student and teachers share the tokenizer and native chat template with thinking disabled. The frozen protocol records model revisions and weight identities.

AdamW configurations use learning rate 2.5×10^{-7} with $(\beta_1, \beta_2) = (0.9, 0.98)$, $\epsilon = 10^{-8}$, no weight decay or schedule, and global norm-1 clipping. Training sampling uses temperature 1 and top- $p = 1$, with a 2,048-token prompt cap and 4,096-token response cap. One fresh response is sampled per prompt. Training uses seeds 42–44 for the I64 GT/DT/DR normalization capability summaries and seed 42 for the other comparisons. Each optimizer update aggregates 64 responses across all domain blocks.

The multi-teacher configurations allocate 16 prompts to each domain per update and use target domain weights 1/4. Single-teacher configurations allocate all 64 prompts to mathematics. Prompts follow a seeded shuffled schedule, repeated after exhaustion. I64 uses the top- k intersection objective in Equation 4 with $k = 64$ from initialization; I16 uses $k = 16$. The supplementary PG/SGD branches use a learning rate of 0.0025 and each complete 500 updates. Hardware and deployment differ from the AdamW runs, so the capability comparison does not establish a runtime ranking.

Training prompts come from the Nemotron-3-Nano-RL-Training-Blend: the DAPO and Skywork mathematics subsets, competition coding with unit tests, instruction following, and STEM multiple-choice questions. The frozen derived four-domain release contains the processed prompts and scoring metadata. A seeded shuffle reserves separate 64-record diagnostic and held-out splits per domain; the remaining training splits contain 21,928 mathematics, 19,041 code, 16,447 instruction-following, and 19,542 science records.

A.2 EVALUATION AND SCORING

Each capability suite contains MATH-500 (500 questions from the MATH benchmark (Hendrycks et al., 2021); dataset release), a fixed 128-question LiveCodeBench v6 subset (Jain et al., 2024), IFBench strict (300 questions) (Pyatkin et al., 2025), and GPQA Diamond (198 questions) (Rein et al., 2023). Mathematics, code, and instruction following use one greedy response per question. GPQA uses four responses per question at temperature 1 and top- $p = 1$. Its score is the mean correctness over all four responses, then over questions: average@4 accuracy. Evaluation permits up to 32,768 response tokens, bounded by the remaining configured 32,768-token context. Each complete suite contains 1,720 responses.

The expanded development grid comprises updates 50/100/250/500 for seven Adam trajectories and 100/250/500 for three SGD trajectories, plus the shared initial evaluation. Appendix H.1 describes a separate additional-domain test. Table 2 supplements the joint endpoints in Table 1 with single-teacher scores and intermediate PG/I64 checkpoints.

The four RL teachers were evaluated separately on their respective domain benchmarks using the same prompt and decoding protocol (Table 3). These scores describe the available supervision on each target domain.

A.3 PAIRED COMPARISONS AND EVIDENCE PROVENANCE

We verify question correspondence and bootstrap the per-question score differences 10,000 times using seed 1042; each GPQA question retains its four-response cluster. Figure 6 shows percentile 95% intervals. These are nominal, unadjusted intervals over evaluation questions, conditional on the

Table 2: Supplementary seed-42 capability scores (%). S denotes mathematics-only training and M denotes joint training; both use DR. Single-teacher trajectories and intermediate joint checkpoints complement the initial and joint endpoint scores in Table 1.

Configuration	Updates	Math	Code	IF	GPQA
S-PG	100	77.40	18.75	18.67	31.82
S-PG	250	73.80	20.31	17.00	35.35
S-PG	500	73.80	19.53	18.00	31.82
S-I64	100	74.60	18.75	17.67	33.21
S-I64	250	74.40	21.09	18.33	32.45
S-I64	500	76.00	17.97	18.67	30.05
M-PG	50	69.60	16.41	18.67	28.03
M-PG	100	72.00	17.19	19.33	31.19
M-PG	250	73.40	16.41	26.00	30.18
M-I64-DR	50	69.60	17.19	19.00	31.69
M-I64-DR	100	71.60	17.97	19.00	32.70
M-I64-DR	250	74.80	17.97	26.00	33.96

Table 3: Domain teacher reference scores (%). Science uses GPQA average@4.

Teacher / target domain	Responses	Score (%)	Truncation (%)
Math	500	75.60	6.80
Code	128	23.44	24.22
If	300	29.00	7.33
Science	792	34.60	0.51

trained checkpoints. Training-seed variation is a separate source of uncertainty. The same procedure supplies the initial-to-checkpoint comparisons in the evidence bundle.

The evidence bundles retain frozen scores, question-level records or verified completion summaries, model and data identities, and the analysis scripts. The original benchmarks and expanded matrix share checkpoints and are not independent inventories. Training-seed variation, question-bootstrap uncertainty, and repeated local response banks are reported separately throughout.

B LENGTH-WEIGHTING DERIVATION

Fix a rollout batch, its valid-token masks, and domain weights λ_d . Any PG coefficients are detached. For $n_d = |\mathcal{B}_d|$, let $\bar{T}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} T_i$ and $\bar{g}_d = n_d^{-1} \sum_{i \in \mathcal{B}_d} g_i$. Define the empirical covariance using the same $1/n_d$ normalization:

$$\text{Cov}_d(T, g) = \frac{1}{n_d} \sum_{i \in \mathcal{B}_d} (T_i - \bar{T}_d)(g_i - \bar{g}_d). \quad (11)$$

Expanding gives $n_d^{-1} \sum_i T_i g_i - \bar{T}_d \bar{g}_d$. Dividing by $\bar{T}_d$ yields Equation 6. GT combines the resulting domain gradients using observed token shares; DT combines them using λ_d ; DR combines $\bar{g}_d$ using λ_d . Multiplying each domain’s globally averaged terms by its target weight divided by its observed token share therefore gives DT exactly.

The derivation is a finite-batch weighted-average identity. Response sampling, masks, and lengths are held fixed. Its population analogue involves an expectation of a ratio; replacing that expectation by a ratio of expectations requires additional assumptions or a large-batch limit.

B.1 FIXED-BATCH GRADIENT COMPARISON

At M-PG checkpoints 100 and 250, three banks (seeds 1042, 1043, and 1044) each contain eight responses per domain. Responses use the training prompt format and 4,096-token response cap. For each bank, we compute each response’s mean I64 gradient once at the saved BF16 model values using FP32 computation, then aggregate those gradients under DR, DT, and GT. All three reductions

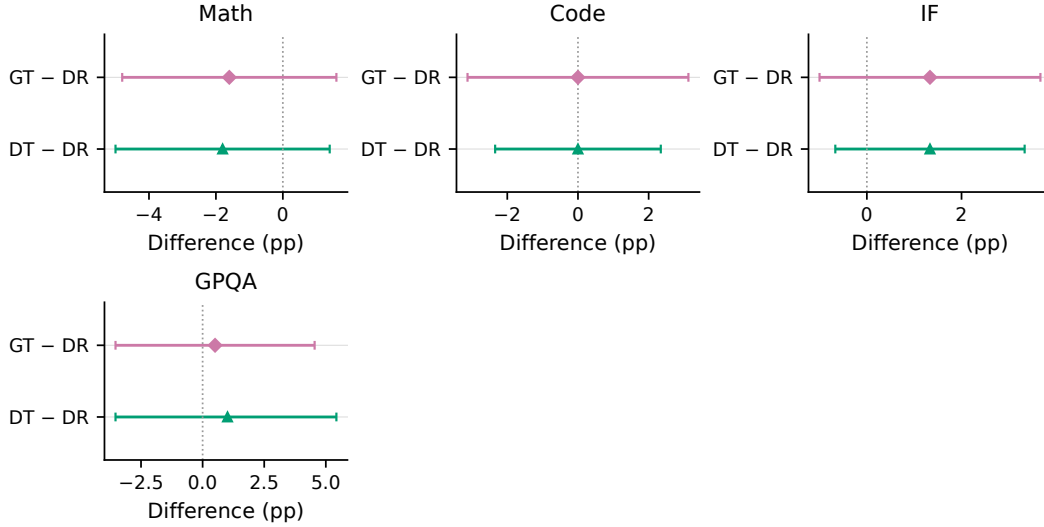


Figure 6: Update-500 I64 capability contrasts against DR, with nominal 95% paired question-bootstrap intervals.

Table 4: Update-500 I64-minus-PG capability differences and nominal 95% paired question-bootstrap intervals (percentage points). All eight intervals include zero.

Setting	Domain	I64 – PG (pp)	95% interval (pp)
Single	Math	+2.20	[-0.80, +5.00]
Single	Code	-1.56	[-3.91, +0.00]
Single	IF	+0.67	[-1.67, +3.00]
Single	GPQA	-1.77	[-5.30, +1.77]
Joint	Math	+2.60	[-0.40, +5.60]
Joint	Code	+0.00	[-2.34, +2.34]
Joint	IF	+0.33	[-2.00, +2.67]
Joint	GPQA	-1.77	[-5.81, +2.40]

use the same responses, token masks, teacher scores, and model coordinates. Student and optimizer snapshots match within each checkpoint.

Table 5 reports every pairwise raw-gradient cosine. The domain-wise covariance decomposition is checked in FP64 over all 24 bank-domain groups; the largest relative residual is 1.53×10^{-15} . This numerical check verifies the aggregation identity. The observed gradient directions establish that the covariance contribution is nonzero on these batches.

The optimizer interventions in Figure 2 reuse these six banks. Each aggregate gradient is globally clipped at norm 1 and applied to the same saved FP32 Adam state for that bank. The saved-state branch retains both moments; the reset- m branch zeros the first moment while retaining the second moment and optimizer clock; reset- (m, v) zeros both moments while retaining the clock. The momentum-free SGD control follows the negative gradient and is rescaled separately for DR, DT, and GT to match that reduction’s saved-state Adam FP32 step norm. Every proposed FP32 step is cast to BF16 before evaluating one held-out response per domain. We report full-vocabulary reverse KL for evaluation; the training-consistent I64 surrogate is retained in the source bundle. These are immediate local interventions, not additional online training runs.

Table 5: Fixed-batch averaging comparison. Entries are raw-gradient cosines; each row uses one shared student and response bank. DR and DT share domain weights and differ only in within-domain response weighting.

Updates	Batch	DR-DT	DR-GT	DT-GT
100	1	0.7838	0.6219	0.9266
100	2	0.9110	0.5180	0.7010
100	3	0.5780	0.6659	0.9745
250	1	0.6240	0.4072	0.8742
250	2	0.7864	0.3302	0.4953
250	3	0.3990	0.3569	0.9612

C PARAMETER AND OPTIMIZER MEASUREMENTS

C.1 COORDINATES AND RECORDED QUANTITIES

The online record distinguishes raw aggregate gradients, clipped gradients, actual FP32 optimizer steps, cumulative FP32 master changes, and cumulative BF16 model changes. Cumulative changes use the same initialization reference. All global statistics use unique model coordinates, excluding vocabulary padding and shared duplicates. Per-layer statistics and absolute-threshold sensitivities are retained in the frozen measurements. The sparsity value is the fraction at or below a threshold; figures report its complement. Energy concentration uses squared Euclidean norm and ranks coordinates by absolute change.

For local calculations, the exported checkpoint contains FP32 master values, BF16 model values, first and second moments, optimizer hyperparameters, and step clocks. Student and teacher values are verified against their saved models in the same precision. Local forward/backward computations evaluate these values in FP32 with TF32 disabled. All proposed steps are computed from the same saved state with a non-fused Adam implementation. The online training optimizer uses its own fused implementation.

We evaluate three local states. *Saved* retains both moments and the step clock. *Reset* m zeros the first moment while retaining the second moment and clock. *Fresh* zeros both moments and resets the clock. Global norm clipping at 1 is applied before the Adam calculation, and weight decay is zero. The plotted controls use two response banks; the zero-gradient control is evaluated once. For the zero-gradient branch, the saved first moment continues to decay and move the master weights. A no-step master-to-BF16 cast gives an exact match to the saved model. Figure 7 shows how absolute thresholds change the measured local activity.

The zero-current-gradient branch changes 0.1403% of BF16 coordinates, compared with 0.1486% for PG. Resetting the first moment gives 0.0504% for PG; resetting both moments and the clock gives 0.5140%. These controls explain why a sparse gradient selection cannot by itself identify the parameters changed by an optimizer step. They supplement the update-sparsity measurements in Sections 4 and 5.

Figure 5(b,c) displays these BF16 controls. The full FP32 and BF16 measurements are retained in the evidence bundle.

C.2 RESPONSE BANKS, OVERLAPS, AND GRADIENT MAGNITUDES

These raw-gradient diagnostics use the M-PG/100 student, whereas Figure 3(b,c) measures teacher-conditioned optimizer steps at M-PG/500. Banks use seeds 42 and 43, one response per domain per bank, a 2,048-token prompt cap, and a 256-token generation cap. All eight responses reach the cap. Four response positions are sampled without replacement per response. Each row’s four prefixes receive equal weight. For routed gradients, each teacher scores its domain row; for common-bank gradients, it scores all four rows with equal domain weights. Crossed comparisons score every domain row with every teacher, keeping the number of prefixes per gradient at four. Crossed and routed/common comparisons share underlying gradients and form paired conditions within banks.

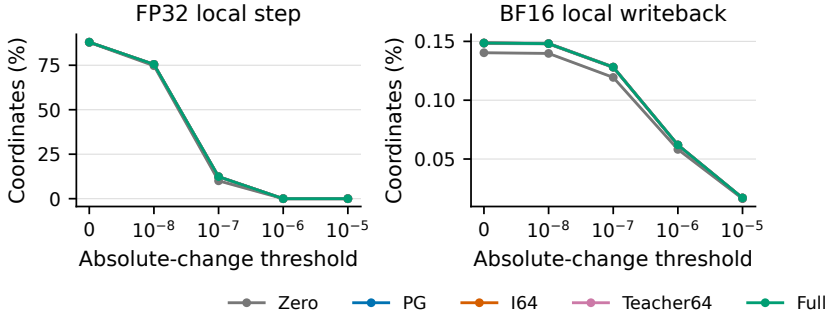


Figure 7: Threshold sensitivity of FP32 steps and BF16 changes. Lines show bank means; shading spans their range.

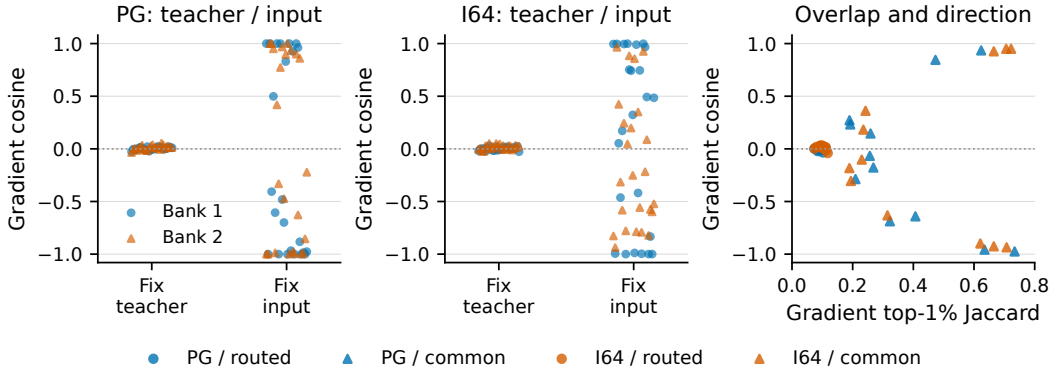


Figure 8: Input dependence of teacher overlap. Left/center: crossed teacher–input comparisons. Right: top-1% gradient Jaccard versus signed cosine for routed and common inputs.

Top- k coordinate sets rank absolute gradient values, using a stable coordinate index to break cut-off ties. Table 7 supplies the routed PG norms; the first bank’s instruction-following gradient has norm 1.53×10^{-9} . Cosine normalizes magnitude, so opposite directions for such a row carry very little absolute gradient mass. Jaccard overlap measures selected locations and cosine measures signed direction. For reference, independent uniform sets of fraction ρ have expected-intersection to expected-union ratio $\rho/(2 - \rho)$, or approximately 0.00503 at $\rho = 1\%$. This reference uses uniform coordinates; layer and weight-magnitude structure remain in the measured gradients.

Applying Equation 8 to the largest-magnitude 1% of raw-gradient coordinates gives mean routed Jaccard overlaps of 0.0951 for PG and 0.0999 for I64 across the two banks and six teacher pairs. Their ranges are 0.0744–0.1077 and 0.0737–0.1178. When every teacher instead scores all four domain responses, the respective means are 0.3808 and 0.4577. The latter comparison changes both input composition and prefix count; the crossed input control below fixes the count.

Holding the teacher fixed while changing the input domain yields mean gradient cosines of 0.0059 for PG and 0.0072 for I64. Holding the input domain fixed while changing teachers produces both aligned and opposed gradients. For routed gradients the mean cosines are -0.0020 and 0.0081 , respectively, despite partial coordinate overlap. Figure 8 supplies this input and direction context for the teacher-overlap results in Section 4.

D VOCABULARY SUPERVISION AND COVERAGE

Four-bank local experiment. The M-PG/100 row of Figure 4(a) uses four response banks with seeds 2042–2045. Each bank contains one response per domain. Generation uses the training prompt format, a 2,048-token prompt cap, and a 4,096-token response cap. The 16 response lengths range

Table 6: Local concentration at M-PG/100, averaged over four matched banks. NZ is the percentage of nonzero coordinates; top-1% is the percentage of squared step norm carried by the largest 1% of coordinates. These are local bank means, not training-seed estimates.

Loss	FP32 NZ	BF16 NZ	FP32 top-1%	BF16 top-1%
PG	87.982	0.149	8.845	100.000
I64	87.989	0.150	8.827	100.000
T64	87.989	0.150	8.828	100.000
Full	87.999	0.150	8.828	100.000

from 81 to 4,096 tokens; two responses reach the cap. Six positions per response give 96 prefixes. PG actions are freshly sampled from the student at the cached prefixes. Every loss uses the same prefixes and domain-response weights. Gradients are globally clipped at norm 1 before applying the same saved Adam state. Table 6 adds the FP32 and energy-concentration measurements for this state.

The local loss and Adam implementations are the same as in the two-bank controls in Appendix C. Appendix H.3 reports the actual I64 intersection coverage across all four student states.

Implementation details. For I64 (Equation 4), candidate sets and teacher scores are fixed at roll-out. Each student forward recomputes current probabilities at the saved candidate positions and refreshes the detached coefficients. PG also refreshes its detached log-ratio coefficients and uses no advantage or policy-ratio clipping.

Teacher-top-64 local reference. The teacher-top-64 diagnostic uses the fixed teacher set T and full-vocabulary probabilities:

$$\ell_{T64} = \sum_{v \in T} \left[p_{\theta}(v) \log \frac{p_{\theta}(v)}{q_d(v)} - p_{\theta}(v) + q_d(v) \right]. \quad (12)$$

Its gradient is $\sum_{v \in T} p_{\theta}(v) \log(p_{\theta}(v)/q_d(v)) \nabla_{\theta} \log p_{\theta}(v)$. I64 instead uses the detached weights in Equation 4, normalized over the original student set S . T64 and full vocabulary are local objective comparisons; online capability curves compare PG, I16, and I64. I16 has no corresponding local geometry measurement.

For a fixed prefix, differentiating Equation 2 gives

$$\nabla_{\theta} \ell_{\text{full}} = \sum_{v \in \mathcal{V}} p_{\theta}(v) \left(\log \frac{p_{\theta}(v)}{q_d(v)} + 1 \right) \nabla_{\theta} \log p_{\theta}(v) \quad (13)$$

$$= \mathbb{E}_{a \sim p_{\theta}} \left[\log \frac{p_{\theta}(a)}{q_d(a)} \nabla_{\theta} \log p_{\theta}(a) \right], \quad (14)$$

using $\sum_v \nabla_{\theta} p_{\theta}(v) = 0$. The identity conditions on prefixes and fixed weights. Original rollout losses additionally use the realized response and its eventual length-dependent weight. Table 7 reports the corresponding coverage for the two-bank controls used to interpret gradient overlap and optimizer history.

E TEACHER DISTRIBUTION DISTANCE

Teacher distance is computed on the full next-token vocabulary at matched prefixes:

$$\text{JS}(q_d, q_e) = \frac{1}{2} D_{\text{KL}}(q_d \| m) + \frac{1}{2} D_{\text{KL}}(q_e \| m), \quad m = \frac{1}{2}(q_d + q_e). \quad (15)$$

We average over the same equally weighted domains and prefixes for every pair. Figure 11(b) pairs this distance with common-input optimizer-step selections at M-PG/500. The earlier M-PG/100 raw-gradient diagnostic contains twelve bank-pair JS measurements, ranging from 0.00151 to 0.02714 nats; its pair-level records remain in the evidence bundle. Both diagnostics share teachers and prefixes within each bank, so teacher-pair observations are dependent.

Table 7: Teacher-top-64 coverage and routed gradient scale in the two-bank controls. Probability mass and student entropy (nats) are prefix averages; displayed 100% values may reflect rounding.

Bank	Domain	Mass in T (%)	Mass in I (%)	Entropy	$\ g_{PG}\ _2$
1	Math	100.0000	100.0000	0.3243	132.6
1	Code	99.9968	99.9965	0.7701	163.8
1	IF	100.0000	100.0000	1.85×10^{-5}	6.13×10^{-9}
1	Science	99.9184	99.9162	0.7405	54.82
2	Math	99.9991	99.9991	0.3221	2.605
2	Code	100.0000	100.0000	0.2962	3.167
2	IF	100.0000	100.0000	0.5004	55.52
2	Science	99.9993	99.9992	0.2442	0.8505

F SUPPORTING VIEWS FOR THE MAIN FIGURES

Capability provenance. Figures 1(c–f) and 9 report means and sample standard deviations over the three experimental seeds 42, 43, and 44, whose provenance is confirmed by the authors. The expanded endpoint matrix and Figure 4(c) use seed-42 scores; these additional contrasts are point estimates rather than tests of training-seed significance.

Teacher-update controls. Figure 3(a) compares DR cumulative changes at update 500. Panels (b,c) use the common M-PG/500 student and optimizer state, separating routed and common inputs. Each of four banks contains six teacher pairs; the displayed bank points and means preserve this grouping. The history-subtracted control uses Equation 9 and reports FP32 update cosines. SGD and reset- m steps are individually norm-matched to saved-state Adam; reset- m retains the second moment and optimizer clock. Teacher-pair observations share teachers and banks and are dependent. Steps use non-fused PyTorch updates on saved-state copies, rather than bitwise replay of the online optimizer. Single-teacher and joint runs differ in per-domain response exposure as specified in Appendix A.

Local supervision controls. Figure 4(a,b) uses four banks at each of four joint-student states: M-PG and M-I64-DR at updates 100 and 500. Losses share prefixes and saved optimizer state within each row, while inputs may differ across rows. Direct T64 step-direction pairs were not recorded, and I16 is evaluated only in the online capability comparison. Panel (c) matches task, setting, and update for S-I64, M-I64, and the added M-I16 branch, each relative to PG in its own teacher setting. Table 6 supplies the FP32 and energy-concentration measurements for the M-PG/100 comparison described in Appendix D.

F.1 ADDITIONAL CONTROL FIGURES

Figure 2 reuses the six fixed banks and online clipping records described in Appendix B.1. Reset- m and reset- (m, v) retain the optimizer clock. Panel (c) averages four domain probe losses within each bank; its error bars are sample standard deviations across the three banks at each checkpoint, not training-seed uncertainty.

Figure 5(a) uses three M-PG/100 rollout banks with a 4,096-token response cap, containing 48 responses and 288 prefixes. Twelve fixed-prefix subbanks each receive two independent PG action draws at counts 1, 16, and 64; the I64 reference is evaluated once per subbank. Mean PG gradient cosines to full vocabulary are 0.5374, 0.8652, and 0.9667; I64 is 0.999988. These are local diagnostic repeats. Panels (b,c) instead use the two 256-token-cap banks from Appendix C; the zero-gradient branch is measured once. Reset- m retains the second moment and clock, whereas fresh Adam resets both moments and the clock. These separate experiments do not form one paired intervention chain.

Cumulative trajectories. Figure 10 extends the update-500 summary in Figure 3(a) to the recorded trajectories.

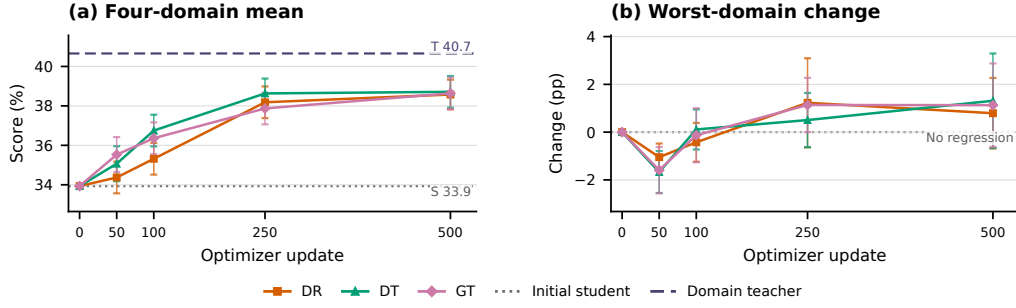


Figure 9: Aggregate GT/DT/DR capability: (a) four-domain mean with student/teacher references; (b) worst-domain change from initialization. Zero denotes no regression. Error bars show mean $\pm$ sample SD across the three reported experimental seeds.

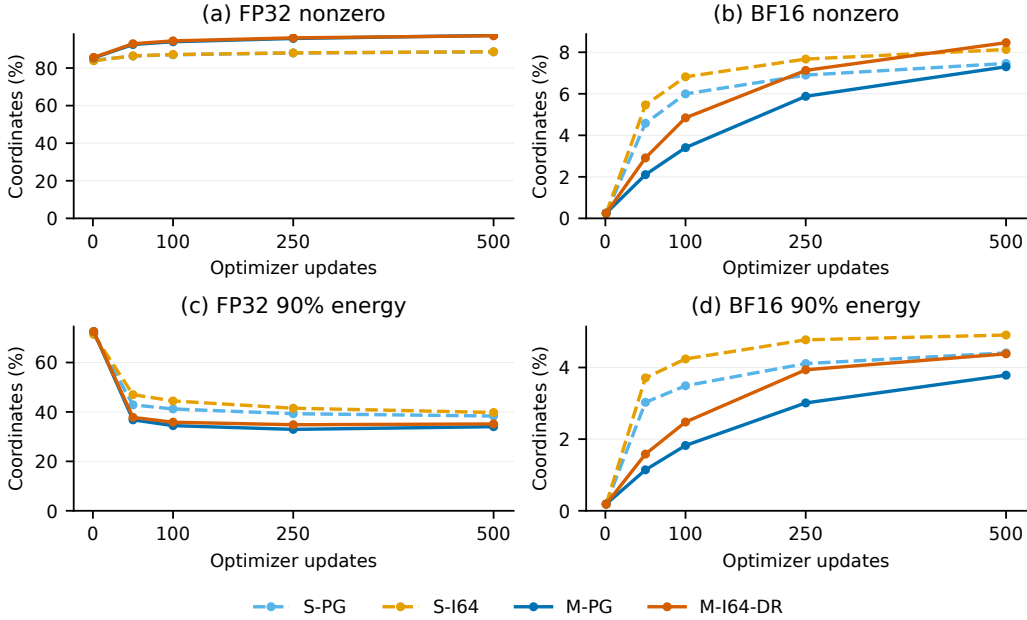


Figure 10: Cumulative PG/I64 changes through update 500 in single-teacher (S) and joint (M) training. (a,b) Nonzero fractions in FP32 and BF16. (c,d) Coordinate fractions carrying 90% of squared change in each precision. Lines connect measured updates 1, 50, 100, 250, and 500; joint I64 uses DR.

G EXPOSURE AND CANDIDATE WEIGHTING

With domain losses averaged separately, prompt quotas and loss weights play different roles: increasing a quota at fixed λ_d changes exposure and estimation variance without directly increasing that domain’s expected gradient weight. The single-teacher runs receive four times as many mathematics prompts per update as joint runs, which limits causal attribution of their capability differences to teacher count alone.

Candidate coverage also changes the scale of vocabulary feedback. For the student candidate set S and teacher intersection I in Equation 4, the retained student weights sum to $r(h) = p_\theta(I | h)/p_\theta(S | h)$. At a fixed prefix with a nonempty intersection and detached coefficients, the I64 gradient is $r(h)$ times the gradient obtained by renormalizing the same weights over I . Thus removing length weighting does not remove variation in feedback scale across prefixes. Monitoring

Table 8: Additional-domain test scores (%). Method rankings vary across domains and can differ from the original benchmarks. Mean weights domains equally.

Configuration	Update	Math	Code	IF	Science	Mean
Initial student	0	42.97	23.44	25.78	62.21	38.60
PG-DR / Adam	500	46.48	26.17	37.50	64.06	43.55
PG-DT / Adam	250	44.53	26.56	36.72	64.75	43.14
PG-DT / Adam	500	44.92	27.34	38.28	63.48	43.51
PG-GT / Adam	500	44.92	28.91	39.06	65.14	44.51
I16-DR / Adam	500	44.92	28.52	40.23	64.55	44.56
I64-DR / Adam	500	46.88	26.17	39.06	65.72	44.46
I64-DT / Adam	500	43.36	27.73	39.06	64.55	43.68
I64-GT / Adam	500	45.31	27.34	41.02	66.21	44.97
PG-DR / SGD	500	46.88	27.73	37.11	62.21	43.48
PG-DT / SGD	500	46.48	28.52	37.11	63.48	43.90
PG-GT / SGD	500	44.53	26.17	36.72	64.45	42.97

this retained mass and empty intersections by domain helps assess whether a fixed candidate size supplies comparable coverage; increasing or renormalizing a small intersection requires validation because it can also amplify unreliable feedback.

H COMPLETED CAPABILITY COMPARISONS AND ADDITIONAL CONTROLS

H.1 EVALUATION SCOPE AND THE INDEPENDENT DOMAIN TEST

The ten seed-42 joint-training trajectories use the development benchmarks and checkpoint grid described in Appendix A. The analyses below add testing on separate questions, frozen checkpoint selection, coverage measurements, and overlap sensitivity.

Table 8 uses a separately frozen additional-domain test with 256 questions per domain, for 1,024 questions and 1,792 responses per model. Mathematics and code are drawn from the Skywork-OR1-RL-Data mathematics and TACO-based code pools; IF uses Nemotron-RL-instruction_following; Science uses the Nemotron-RL-knowledge-mcqa validation split restricted by a fixed natural-science keyword filter. Questions are the first eligible unique candidates in frozen source order, selected without model scores. Normalized exact and lexical near-duplicate checks exclude the original training bundle and known evaluation pools. These checks do not establish exhaustive semantic decontamination or absence of pretraining exposure.

The additional test uses math answer verification, code unit tests, instruction-following constraints, and science multiple-choice scoring. Mathematics, code, and IF use greedy decoding; Science uses four responses per question at temperature 1. The generation context budget is 32,768 tokens, with thinking disabled. The source distributions and the code/IF scorers differ from the original benchmark protocol, so the two tables are reported separately. The four-domain mean is an equal-weight descriptive aggregate. The available question-paired bootstrap intervals describe finite-test uncertainty, not variability across independent training seeds.

H.2 FROZEN CHECKPOINT-SELECTION OUTCOMES

The three selection rules in Section 6 are applied to the development grids in Appendix A. Selections were frozen before independent test inference and are not recomputed using the test or overlap-sensitivity results. The additional PG-DT/Adam evaluation at update 250 is included because both development selection rules chose it.

All selected models satisfy the retention and useful-gain conditions as test point estimates; this is not a simultaneous confidence guarantee. PG-DR/Adam nevertheless has two development checkpoints outside the tolerance: GPQA falls from 32.197% initially to 28.030% at update 50 and 30.177% at update 250, before reaching 34.975% at update 500. Endpoint retention therefore does not imply retention throughout training.

Table 9: Frozen checkpoint choices. Max utility and retention select the same checkpoint in every trajectory. Test Δ is the selected-checkpoint mean minus the endpoint mean in percentage points.

Trajectory	Endpoint	Max utility	Retention	Test Δ (pp)
Other nine trajectories	500	500	500	0.000
PG-DT / Adam	500	250	250	-0.366

Table 10: I64 candidate intersections retain high student probability mass. Mass is in percent; means and minima are over sampled prefixes. $r = p_\theta(I | h)/p_\theta(S | h)$ is the retained weight ratio.

Student state	$p(I)$: mean	$p(I)$: min	$ I $: mean	$ I $: min	r : mean	r : min
M-PG/100	99.9141	96.5482	58.96	43	0.9999413	0.9977714
M-PG/500	99.9952	99.7825	61.59	50	0.9999975	0.9998511
M-I64-DR/100	99.9564	97.5664	60.08	36	0.9997331	0.9757993
M-I64-DR/500	99.9826	99.3722	61.75	51	0.9999912	0.9997005

H.3 ACTUAL INTERSECTION COVERAGE

For each of four student states (M-PG and M-I64-DR at updates 100 and 500), four banks provide one response per domain and six sampled positions per response: 96 prefixes per state. Each response is scored by its domain teacher. We measure $p_\theta(I | h)$ for the actual intersection $I = S \cap T$ and the retained fraction $r = p_\theta(I | h)/p_\theta(S | h)$.

Table 10 measures the student probability mass in the actual student–teacher candidate intersection, rather than in the teacher candidate set alone. Across 384 routed prefixes, the minimum intersection mass is 96.55%, and none has an empty intersection. This high-coverage regime helps delimit the local comparisons with full-vocabulary feedback. It does not establish the same behavior for low-coverage prefixes, other teacher configurations, or the full online I16 training trajectory. The four banks are local sampling repeats, not independent training seeds; inputs can differ across student states.

H.4 SENSITIVITY TO KNOWN MATHEMATICS OVERLAP

The frozen Qwen training pool contains five records matching four known MATH-500 underlying problems, including wording or answer-form variants. The affected MATH test identifiers are counting-and-probability problem 430 and intermediate-algebra problems 1849, 2146, and 582. Their presence in the training pool does not establish that each was sampled by a particular run or caused a performance increase. We remove these four fixed evaluation questions from the existing per-question scores as an offline sensitivity analysis, retaining the original 500-question results as the main protocol.

The ordering by four-domain mean is unchanged across the twelve reported checkpoints. MATH-496 denotes the retained subset, not a new official benchmark. The frozen checkpoint selections are retained. This comparison neither retrains on a filtered pool nor constitutes an exhaustive overlap or contamination audit.

H.5 TEACHER-UPDATE OVERLAP AND DISTRIBUTIONAL SIMILARITY

Figure 11 complements the main history-subtracted analysis using the M-PG/500 state and controls described in Appendix F. It compares PG/I64 under saved-state Adam, reset- m , and norm-matched SGD. Common inputs hold student prefixes fixed across teachers; routed inputs change the teacher and its input domain together. Each of four banks contains six teacher pairs; observations share teachers and banks and are dependent.

Table 11: Offline sensitivity to excluding four known overlapping MATH-500 questions. The four-domain mean ranking is unchanged. Mean (496) replaces only the mathematics score with that on the retained subset.

Configuration	Update	MATH-500	MATH-496	Mean (500)	Mean (496)
Initial student	0	68.80	69.15	33.93	34.02
PG-DR / Adam	500	73.20	73.79	38.29	38.43
PG-DT / Adam	250	77.00	77.42	38.78	38.88
PG-DT / Adam	500	73.20	73.59	38.03	38.13
PG-GT / Adam	500	76.20	76.81	38.36	38.52
I16-DR / Adam	500	75.40	76.01	38.65	38.80
I64-DR / Adam	500	75.80	76.41	38.58	38.73
I64-DT / Adam	500	74.00	74.40	38.71	38.81
I64-GT / Adam	500	74.20	74.60	38.64	38.74
PG-DR / SGD	500	76.40	77.02	39.46	39.62
PG-DT / SGD	500	75.20	75.81	39.14	39.29
PG-GT / SGD	500	75.00	75.60	38.39	38.55

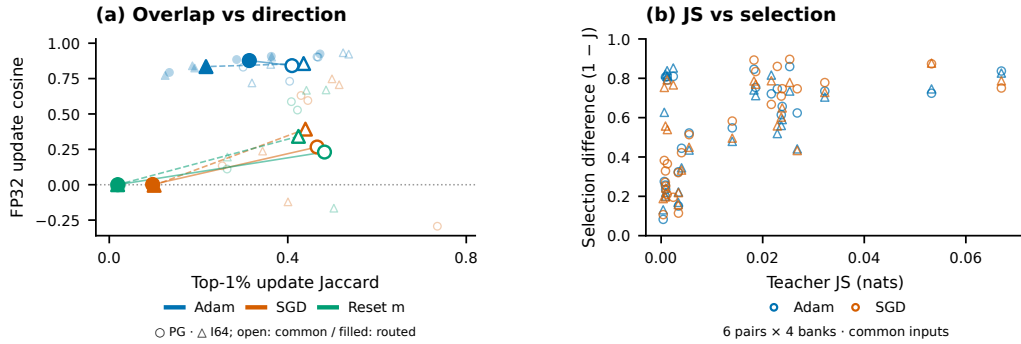


Figure 11: Teacher-update overlap and distributional similarity. (a) Top-1% FP32 update-set Jaccard versus cosine; small points summarize banks and large points average banks. (b) Common-input teacher JS versus 1-Jaccard; each point is a teacher pair within a bank. These geometric associations do not measure cross-domain capability harm.